CSST - Complication Signal Surveillance Template

Review identification, purpose and sources

SIMULATED · v2.0

Study / protocol GLENOS-01 / GS-2026-01	**CIP version** CIP v4.2	**Site / code** Riverside Orthopedic Center / 014	**CSST review ID / revision** CSST-014-02 / r1
Review type ☐ Scheduled ☑ Triggered	**Review period / data cut-off** 1-31 Oct 2026; cut-off 31 Oct	**Device / procedure reviewed** GLENOS reverse shoulder, all study procedures	**Previous CSST review ID** CSST-014-01 (30 Sep 2026)
SCML snapshot ID SCML-014, snapshot 31 Oct 2026	**Analysis file (if used)** None: counted on this form	**Prepared by / role** Maria Silva / Study Coordinator	**Controlled location** ISF tab 6 (Safety) / CSST-014-02

A Purpose and use

The CSST is the site's review of its complications taken together. A single event is assessed in the SART and indexed in the SCML; the CSST looks at all of them at once, on the site's own schedule and after a trigger, to see whether events that each seemed explicable share a cause: the same component, the same surgeon, the same moment of follow-up or the same kind of case. A concentration of events is a question to test, not a conclusion about a device or a surgeon.

Unit of analysis	Decision supported	Inputs	Outputs
One review of the site's complications for one period, or one review opened by a trigger.	Whether the data show no material pattern, a pattern to watch more closely, a signal to send on, or a result limited by the data.	SCML events, ICDL procedures and context, PFVR follow-up reached, SLCT surgeon sequence, SART event status and the previous CSST.	Counts by time point (C) Triggers met (D) Signals assessed (F) Routes to the receiving records (H)

B Sources and their limitations

With limitation means the review goes ahead: name the records that are missing or incomplete. Not available means the counts that need that source cannot be made in this review.

Source	Used for	ID · extraction date	Status	Limitation (records missing or incomplete)
SCML snapshot	Events and their categories	SCML-014 snapshot · 31 Oct	☑ Complete ☐ With limitation ☐ Not available	
ICDL procedure records	Procedures, components, configuration, context	ICDL-014-001 to -008 · 31 Oct	☐ Complete ☑ With limitation ☐ Not available	ICDL-014-005: liner lot missing; read from the explant label
PFVR / visit schedule	Participants who reached each follow-up time point	PFVR export · 31 Oct	☐ Complete ☑ With limitation ☐ Not available	014-0003: 6-week visit on day 52 (outside window)
SLCT	Surgeon codes and case sequence	SLCT-014 · 31 Oct	☑ Complete ☐ With limitation ☐ Not available	
SART	Event type, seriousness and status	SART export · 31 Oct	☑ Complete ☐ With limitation ☐ Not available	
Previous CSST review	Signals to follow up in G	CSST-014-01 · 30 Sep	☑ Complete ☐ With limitation ☐ Not available	

C1 Cases and follow-up reached

These numbers are the denominators for C2 and C3. Procedures and surgeons come from the ICDL and the SLCT, enrollment from the PSEAR, follow-up from the PFVR. The time points match the columns of C2. Adapt the time points to the CIP visit schedule.

Count	This period (n)	Previous period (n)	Change (%)	Cumulative (N)
Participants enrolled	4	4	0%	8
Investigational procedures performed	4	4	0%	8
Surgeons operating	3	2	+50%	3
Surgeons with a first study case in this period	1	2	-50%	3
Procedures within a surgeon's first 5 study cases	3	4	-25%	7
Revision procedures (component exchanged or removed)	1	0	-	1
Explanted components returned	0 (1 pending)	0	-	0
Follow-up visits due	9	4	+125%	13
Follow-up visits done in window	7	4	+75%	11
Follow-up visits missed	1	0	-	1
Participants who reached each time point (N for the columns of C2)				
Operated (intraoperative)	4	4	0%	8
Reached discharge	4	4	0%	8
Reached first postoperative visit	4	4	0%	8
Reached 6 weeks	3	2	+50%	5
Reached 3 months	2	0	-	2
Reached 6 months	0	0	-	0
Reached 12 months	0	0	-	0
Reached 24 months	0	0	-	0
Participants withdrawn or lost to follow-up	0	0	-	0

C2 Complications by follow-up time point

Count each event once, in the time point where it occurred, against the N of that column. Reading across a row shows whether a problem appears early or grows with time.

Event (SCML category)	Intraoperative (n / N) %	Before discharge (n / N) %	First postop visit (n / N) %	6 weeks (n / N) %	3 months (n / N) %	6 months (n / N) %	12 months (n / N) %	24 months (n / N) %	Total (n)	Grade IIIa or higher (n)	New this period
N for this column (from C1)	8	8	8	5	2	0	0	0			
Instability / dislocation											
Dislocation				(1 / 5) 20%					1	1	1
Subluxation or instability symptoms					(1 / 2) 50%				1	0	1
Fracture											
Intraoperative fracture									0	0	0
Periprosthetic fracture									0	0	0
Acromion or scapular spine fracture									0	0	0
Infection / wound											
Superficial wound problem		(1 / 8) 12.5%							1	0	0
Deep or periprosthetic infection									0	0	0
Hematoma		(1 / 8) 12.5%							1	0	1
Neurologic / vascular											
Nerve injury (axillary nerve, brachial plexus)			(1 / 8) 12.5%						1	0	0
Vascular injury									0	0	0
Device performance											
Component loosening									0	0	0
Component disassociation (liner, glenosphere)				(1 / 5) 20%					1	1	1
Implant breakage or wear									0	0	0
Radiographic change (notching, lucency, migration)									0	0	0

C2 Complications by follow-up time point (continued)

Event (SCML category)	Intraoperative (n / N) %	Before discharge (n / N) %	First postop visit (n / N) %	6 weeks (n / N) %	3 months (n / N) %	6 months (n / N) %	12 months (n / N) %	24 months (n / N) %	Total (n)	Grade IIIa or higher (n)	New this period
N for this column (from C1)	8	8	8	5	2	0	0	0			
Function and other											
Stiffness or loss of motion									0	0	0
Persistent pain									0	0	0
Thromboembolic event									0	0	0
Other (SCML code): _____									0	0	0

Overview

Measure	This period (n)	Previous period (n)	Change (%)	Cumulative (n / N) %	Cumulative at previous review (n / N) %
Participants with at least one complication	3	2	+50%	(5 / 8) 62.5%	(2 / 4) 50%
Participants with a complication of grade IIIa or higher	1	0	-	(1 / 8) 12.5%	(0 / 4) 0%
Participants with more than one complication	1	0	-	(1 / 8) 12.5%	(0 / 4) 0%
Reoperations	1	0	-	(1 / 8) 12.5%	(0 / 4) 0%
Revisions (component exchanged or removed)	1	0	-	(1 / 8) 12.5%	(0 / 4) 0%
Serious events (SAE / SADE)	1	0	-	(1 / 8) 12.5%	(0 / 4) 0%
Device deficiencies (all; counted apart from complications)	1	0	-	(1 / 8) 12.5%	(0 / 4) 0%

C2 Complications by Clavien-Dindo grade and time point

Grades come from the SCML, which copies them from the SART. Count each complication once, in the column of its time point. Device deficiencies without an adverse event are not graded.

Grade	Intraoperative (n / N) %	Before discharge (n / N) %	First postop visit (n / N) %	6 weeks (n / N) %	3 months (n / N) %	6 months (n / N) %	12 months (n / N) %	24 months (n / N) %	Total (n)	New this period
N for this column (from C1)	8	8	8	5	2	0	0	0		
I · No treatment beyond simple care		(1 / 8) 12.5%	(1 / 8) 12.5%		(1 / 2) 50%				3	2
II · Drug treatment, transfusion		(1 / 8) 12.5%							1	0
IIIa · Intervention without general anesthesia									0	0
IIIb · Intervention under general anesthesia				(2 / 5) 40%					2	2
IVa · Life-threatening, single organ									0	0
IVb · Life-threatening, multiorgan									0	0
V · Death									0	0
Disability at discharge (suffix d)									0	0

C3 Procedure context

For each marker, compare the complication rate in procedures with it and without it. Context never changes the counts in C2; it helps explain them.

Context marker	Procedures with it (n / N) %	With it: had a complication (n / n) %	Without it: had a complication (n / n) %	Source
Unplanned configuration change	(1 / 8) 12.5%	(1 / 1) 100%	(4 / 7) 57%	ICDL-014-005
Component substitution	(0 / 8) 0%	-	(5 / 8) 62.5%	ICDL
Technical deviation from the surgical technique	(0 / 8) 0%	-	(5 / 8) 62.5%	ICDL
Intraoperative difficulty (bone quality, deformity, revision case)	(2 / 8) 25%	(1 / 2) 50%	(4 / 6) 67%	ICDL-014-002, -007
Senior support or proctoring used	(2 / 8) 25%	(1 / 2) 50%	(4 / 6) 67%	ICDL / SLCT
Surgeon within first ___ study cases	(7 / 8) 87.5%	(5 / 7) 71%	(0 / 1) 0%	SLCT
Operation outside normal hours	(1 / 8) 12.5%	(1 / 1) 100%	(4 / 7) 57%	ICDL-014-005
Instrument or loaner set problem	(1 / 8) 12.5%	(1 / 1) 100%	(4 / 7) 57%	ICDL-014-002
Gap in the operative record	(1 / 8) 12.5%	(1 / 1) 100%	(4 / 7) 57%	ICDL-014-005
Other: _____				

Links between context and complications

Context marker	Procedures (ICDL IDs)	Complication found (C2 event · SCML ID)	Comment
Unplanned configuration change	ICDL-014-005	Dislocation and liner disassociation (SCML-014-07, -08)	Planned +6 liner not in the set; +3 used. Same case had the weekend list and the missing lot.
Instrument or loaner set problem	ICDL-014-002	Subluxation symptoms at 3 months (SCML-014-09)	Trial liner missing from the loaner set; liner chosen without a trial reduction.

E Comparison by surgeon, component and case type

Grades follow Clavien-Dindo (SCML). A group stands out when its share of grade IIIa–V complications is clearly larger than its share of procedures. Use the surgeon codes of the SLCT and the component, lot and configuration recorded in the ICDL.

E1 · By surgeon (SLCT code)

Surgeon	Procedures (n) % of all	Grade I–II (n / n) %	Grade IIIa–V (n / n) %	Share of all IIIa–V %	IIIa–V in first ___ cases (n / n) %	Note
S-01	(3) 37.5%	(1 / 3) 33%	(0 / 3) 0%	0%	(0 / 3) 0%	PI; experienced
S-02	(3) 37.5%	(2 / 3) 67%	(0 / 3) 0%	0%	(0 / 3) 0%	Sub-I; grade I and II only
S-03	(2) 25%	(1 / 2) 50%	(1 / 2) 50%	100%	(1 / 2) 50%	New in Oct; the only IIIb

E2 · By component, lot or configuration (ICDL)

Component · lot · configuration	Procedures (n) % of all	Grade I–II (n / n) %	Grade IIIa–V (n / n) %	Share of all IIIa–V %	Note
Liner 36 +3 · lot GS-L36-2607	(2) 25%	(1 / 2) 50%	(1 / 2) 50%	100%	Includes ICDL-014-005
Liner 36 +6	(4) 50%	(2 / 4) 50%	(0 / 4) 0%	0%	
Glenosphere 36 · all cases	(8) 100%	(4 / 8) 50%	(1 / 8) 12.5%	100%	No comparison possible

E3 · By case type

Case type (indication, primary or revision)	Procedures (n) % of all	Grade I–II (n / n) %	Grade IIIa–V (n / n) %	Share of all IIIa–V %	Note
Cuff tear arthropathy, primary	(5) 62.5%	(2 / 5) 40%	(0 / 5) 0%	0%	
Fracture sequelae, primary	(2) 25%	(1 / 2) 50%	(1 / 2) 50%	100%	Both instability events
Massive cuff tear with arthritis	(1) 12.5%	(1 / 1) 100%	(0 / 1) 0%	0%	

D Signal triggers

Apply each trigger and write the value found. A trigger met opens a signal in F.

ID	Trigger	What is compared · where	Agreed limit	Source of the limit	Value found (n / N) %	Met?
T1	**Rate above the agreed limit**	Any category at any time point; one limit for all grades and a lower one for grade IIIa–V (C2).	Instability: all grades > 10%; IIIa–V > 5%	Site rule, 20 Aug 2026 (POT)	All: (1 / 2) 50% at 3 mo; IIIa–V: (1 / 5) 20% at 6 wk	☑ Yes ☐ No ☐ Not assessable
T2	**Rate rising with time**	The same category grows from earlier to later time points (C2 row).	Higher at a later time point, 2+ events	Site rule, 20 Aug 2026	Instability 0% → 20% → 50%	☑ Yes ☐ No ☐ Not assessable
T3	**Rise since the previous review**	Same category and time point, compared with the previous CSST.	Double the previous review	Site rule, 20 Aug 2026	Instability 0 → 2 events	☑ Yes ☐ No ☐ Not assessable
T4	**New category**	A category with no earlier events now has events (C2, New this period).	Any first event in a category	Site rule, 20 Aug 2026	Instability; device performance	☑ Yes ☐ No ☐ Not assessable
T5	**Surgeon share**	A surgeon's share of grade IIIa–V complications exceeds their share of procedures (E1).	IIIa–V share 2x share of procedures, 2+ events	Site rule, 20 Aug 2026	S-03: 100% vs 25%, but 1 event	☐ Yes ☑ No ☐ Not assessable
T6	**Component or configuration share**	Same comparison for grade IIIa–V by component, lot or configuration (E2).	Same rule as T5	Site rule, 20 Aug 2026	Liner +3: 100% vs 25%, but 1 event	☐ Yes ☑ No ☐ Not assessable
T7	**Early learning cases**	Grade IIIa–V rate in a surgeon's first cases above their later cases (E1, C1).	Assessable from 6 cases per surgeon	Sponsor guidance GS-RA-01	All surgeons have 3 cases or fewer	☐ Yes ☐ No ☑ Not assessable
T8	**Context association**	Rate with a context marker above the rate without it (C3).	Rate with marker 2x without, 2+ events	Site rule, 20 Aug 2026	Each marker has 1 event only	☐ Yes ☑ No ☐ Not assessable
T9	**Serious events grouped**	Serious events in one category within a short window (SART, C2).	2 serious events, same category, 90 days	CIP 11.6	1 SADE (dislocation)	☐ Yes ☑ No ☐ Not assessable
T10	**Device deficiencies of one type**	Repeated deficiencies of the same type or component (C2, SART).	2 deficiencies of the same type	CIP 11.6	1 (liner disassociation)	☐ Yes ☑ No ☐ Not assessable
T11	**Same severe complication repeated**	The same complication of grade IIIa–V in two or more participants (C2).	Same IIIa–V complication in 2 participants	Site rule, 20 Aug 2026	Dislocation IIIb: 1 participant	☐ Yes ☑ No ☐ Not assessable

These triggers are examples. Before the first inclusion, review them and set each limit with the site's clinical team and, where it applies, the sponsor; write the agreed limit and its source (CIP, sponsor guidance, site procedure). A limit changed later applies only to later reviews.

F Assessment of each signal

One row per trigger met. Describe the pattern and the explanations still to test; do not conclude the cause. Set how the next review will show whether the response worked.

Signal ID	Trigger (D) · events (SCML IDs)	Pattern described	Explanations to test	Disposition	What will show the response worked
S-01	T1-T4 · SCML-014-07, -08, -09	Two instability events in 8 procedures, both after discharge and rising at later visits. One grade IIIb (dislocation with liner disassociation, revised) after an unplanned +3 liner; one grade I (subluxation) after a loaner set without a trial liner. The only IIIb is in S-03 and a fracture-sequelae case.	☑ Device, component or lot ☑ Procedure or technique ☑ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☑ Sent on (see H) ☐ Inconclusive: data limitation	No new instability in the next 8 procedures; full liner range and trial liners in every set (ICDL); CSST monthly for 3 months.
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	
			☐ Device, component or lot ☐ Procedure or technique ☐ Surgeon or learning curve ☐ Case complexity or participant ☐ Data or documentation ☐ Local process ☐ Undetermined	☐ No material pattern ☐ Closer surveillance ☐ Sent on (see H) ☐ Inconclusive: data limitation	

CSST - Complication Signal Surveillance Template

Earlier signals and routes

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G Signals from earlier reviews

Check each earlier signal against the criterion set for it in F.

Signal ID · review	Criterion set then	Result in this review	Outcome	Next step
S-01 · CSST-014-01	No further nerve injury in the next 4 procedures	0 nerve injuries in 4 procedures (Oct)	☑ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	Close; keep in the routine count
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	
			☐ Effective ☐ Partly effective ☐ Not effective ☐ Not assessable	

H Where each signal goes

The action is followed in the receiving record; this review records only that it was sent.

Route	Question for this review	Signal ID (F)	Receiving record · date sent	Status
SLCT	Could surgeon-dependent variation explain the pattern?	S-01	SLCT S-03 review, 5 Nov 2026	☑ Sent ☐ Not required ☐ Pending
Sponsor device-deficiency route	Does the signal point to a device, component, configuration or lot?	S-01	GS-DD-014-0012, 28 Oct 2026	☑ Sent ☐ Not required ☐ Pending
SART	Are the individual events complete and routed through the safety pathway?	S-01	SART-014-005; SART-014-006 opened, 30 Oct	☑ Sent ☐ Not required ☐ Pending
SRR	Does the signal show a local process weakness to enter as a risk?	S-01	SRR-002 reopened; SRR-006, 31 Oct	☑ Sent ☐ Not required ☐ Pending
CAPA log	Did a first response fail, or does a process failure recur?	-	First occurrence; no failed response yet	☐ Sent ☑ Not required ☐ Pending
POT	Does the PI need to restrict, supervise more, continue or escalate?	S-01	POT review 3 Nov: S-03 proctored for 3 cases	☑ Sent ☐ Not required ☐ Pending
PIR / IMC	Should the result inform an indicator or an internal check?	S-01	IMP v1.1 CH-03: CSST monthly	☑ Sent ☐ Not required ☐ Pending

Conclusion and sign-off

The conclusion reflects the most serious open signal.

Review conclusion	☐ No material pattern ☐ Closer surveillance ☑ Signal sent on ☐ Inconclusive: data limitation
Next review · date and what will be checked again	30 Nov 2026, then monthly to January: instability by time point, liner +3 cases, S-03 cases, liner range in every set.

Limitations and rationale

Very small numbers: 8 procedures, 2 participants at 3 months; the percentages move a lot with one event and no test of difference is attempted. The explanted liner has not yet been returned to the sponsor, so the device route has no result yet. ICDL-014-005 lot was read from the explant label.

Rationale: three triggers on instability point to the same few cases; device, technique and the new surgeon cannot be separated yet. Each route was sent so that each owner can test its own explanation.

Reviewer

Name / role: Maria Silva / Study Coordinator

Signature / date:

M. Silva 31 Oct 2026

PI / accountable approver

Name / role: Dr Alex Morgan / PI

Signature / date:

A. Morgan 3 Nov 2026